Ranga Reddy

Lead Data Engineer | Data Platform Architect

Bangalore, India | +91 9986183183 | rangareddy.avula@gmail.com

linkedin.com/in/rangareddy1988 | rangareddy.github.io | github.com/rangareddy

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PROFESSIONAL SUMMARY

Lead Data Engineer and Data Platform Architect with 14+ years of experience building, scaling and optimizing petabyte-scale data platforms on AWS, Azure and GCP. Specialist in Apache Spark (Scala, Python, SQL) and open table formats (Apache Iceberg, Apache Hudi, Delta Lake), with 70 merged PRs in Apache Hudi and 9 in Apache XTable. Track record of cutting cloud costs by up to 40%, MTTR by 45% and incident detection time by 60% for enterprise and Fortune 500 customers. Uses AI-assisted engineering (Claude, GitHub Copilot, Cursor) and builds AI diagnostic agents to improve reliability and developer productivity.

TECHNICAL SKILLS

Architecture & Patterns: Data Lakehouse Architecture, Real-Time Streaming, Batch Processing, Multi-Writer Concurrency, Performance Tuning, Cost Optimization.

Lakehouse & Table Formats: Apache Iceberg, Apache Hudi, Delta Lake, Apache XTable, Hive Metastore.

Distributed Compute & Query Engines: Apache Spark (Core, SQL, DataFrames, Structured Streaming), Trino/Presto, AWS Athena, AWS Glue, BigQuery, Databricks.

Cloud & Orchestration: AWS (S3, EMR, Glue, Athena), GCP (Dataproc, GCS, BigQuery), MinIO, Docker, Kubernetes, CI/CD (Jenkins, Git), Lakeflow Jobs, Apache Airflow.

Languages & Frameworks: Scala, Python, Java, SQL, Spring, Hibernate, JUnit.

Streaming & Messaging: Apache Kafka, Flink, Hudi Streamer, Apache Storm, Spark Structured Streaming.

File Formats: CSV, Parquet, ORC, Avro, JSON.

Databases: PostgreSQL, MySQL, Oracle, Apache HBase.

Data Modeling & Architecture: Medallion Architecture (Bronze/Silver/Gold), Fact & Dimension Modeling, Star Schema, SCD Types, CDC, Partitioning, Clustering, Z-Ordering.

Data Governance, Quality & Observability: Governance Frameworks, Data Quality Rules & Validation, Data Lineage & Cataloging, Metadata Management, Schema Evolution & Registry, Freshness Monitoring, Reconciliation, Root Cause Analysis, Data Profiling.

AI-Assisted Engineering: GitHub Copilot, Anthropic Claude, Cursor IDE, Databricks Genie, Gemini, Prompt Engineering, Automated Test Synthesis.

PROFESSIONAL EXPERIENCE

Onehouse, Bangalore

Open-Source Engineer | Solution Architect | Sep 2024 – Sep 2026

- Built cost-analysis tooling that turns table metadata, query logs and statistics into actionable metrics, flagging small files, orphaned data and stale snapshots. Customers cut storage and compute spend by up to 40%.
- Built an AI Spark Diagnostic Agent that parses driver/executor logs and flags stage bottlenecks and skew, cutting mean time to detect production issues by ~60% (hours to under 30 min).
- Ran 10TB TPC-DS benchmarks comparing Onehouse Quanton and Databricks Photon on Iceberg and Delta Lake; results drove compute-sizing guidance for enterprise accounts.
- Delivered 70 merged PRs in Apache Hudi and 9 in Apache XTable across indexing, compaction, clustering, the timeline, the metadata table, record merger and multi-writer concurrency, including fixes for edge-case correctness bugs.
- Lead escalation engineer for community and enterprise Hudi issues across Spark, Flink, Trino and Presto; built correctness and performance suites used as a release gate.
- Guided enterprise upgrades from Hudi 0.8 to 0.15, tuning clustering, compaction and file sizing to recover query performance.

Cloudera, Bangalore

Staff Software Engineer (Spark & Iceberg Backline) | Mar 2020 – Aug 2024

- Served as the final escalation authority for Cloudera's global support organization on Spark and Iceberg production outages, handling P0/P1 incidents for Fortune 500 customers.
- Built an internal log-parsing tool that ingests Spark event logs, application logs, and Iceberg metadata files, then surfaces stage-level bottlenecks and table-level inefficiencies. Adopted across the global support team; cut average MTTR by 45%.
- Created and taught "Spark Hogwarts," a hands-on curriculum on Spark internals, memory management, and Iceberg mechanics. Upskilled 100+ engineers and contributed to a 35% drop in operational escalations.
- Diagnosed S3 and GCS object storage bottlenecks affecting high-throughput Spark and Iceberg jobs — including file listing inefficiency and metadata commit contention. Fixes shipped as config guidance and backported patches.

Dell Technologies, Bangalore

Big Data Engineer | Nov 2015 – Mar 2020

- Built the ingestion, transformation, and serving layer for SupportAssist Intelligence, moving multi-terabyte unstructured log data through Spark, Kafka, Storm, and HBase into customer-facing dashboards.
- Architected ingestion and CDC modules in PySpark and Spark SQL to parse complex JSON payloads — handling schema drift and malformed records without downstream breaks.
- Designed conceptual, logical, and physical data models for analytics, reporting, and operational systems, including the OMNEO UI.
- Co-led a production HDP 2.3 → 2.5 upgrade across 40 nodes with no unplanned downtime.

Mindtree Pvt. Ltd, Bangalore

Senior Software Engineer | Aug 2014 – Nov 2015

- Built backend components for a meeting-services application in Java, Spring, Hibernate, and MySQL — controller, service, and data-access layers.
- Worked directly with client teams to scope and deliver change requests against a live production system.

ITApp Software Pvt. Ltd, Bangalore

Junior Software Engineer | Oct 2012 – Jul 2014

- Developed Spring MVC modules and REST workflows for a multi-tenant cloud management platform.
- Implemented provisioning and resource lifecycle components backed by Hibernate and MySQL.

OPEN-SOURCE CONTRIBUTIONS & ACHIEVEMENTS

Apache Hudi & XTable Contributor - Active upstream contributor focused on core lakehouse infrastructure - delivering optimizations across indexing mechanics, data compaction strategies, table clustering, and cross-format interoperability layers.

Cloudera Community Spotlight - Recognized as a "Rising Star" member with 72 badges and 10+ accepted solutions.

Publish deep-dive articles on Hudi and Iceberg internals (write path, indexing, COW vs MOR, Iceberg 1.12 upgrade guide) at rangareddy.github.io, reaching 10K+ impressions on LinkedIn.

CERTIFICATIONS:

- Databricks Certified Data Engineer Associate (In Progress, expected 2026)

EDUCATION

Master of Computer Applications (MCA) – Sri Krishnadevaraya University, Anantapur | 2008 - 2011 | Score: 80%.